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CAPTURING HIGH FREQUENCY FEATURES USING HF-BOOST BASED SPECTRALLY ADAPTIVE PHYSICS-INFORMED NEURAL NETWORKS

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Abstract. Neural networks are known to exhibit spectral bias, often learning low-frequency components faster than high-frequency components [16, 21]. This limitation is especially important in scientific machine learning, where sharp gradients, boundary layers, shocks, and oscillatory solutions may carry substantial high-frequency content. We propose HF-Boost, a spectrally adaptive physics-informed neural network (PINN) training strategy that monitors the PDE residual in Fourier bands and increases the PDE-loss weight when high-frequency residual energy is prominent. On Burgers' equation and additional benchmark PDEs, the method improves high-frequency-region accuracy relative to fixed-weight PINN baselines.

1. Introduction

Many complex physical phenomena are modeled by partial differential equations (PDEs), and closed-form analytical solutions are unavailable for most realistic geometries, coefficients, and boundary conditions [4]. Classical numerical methods such as finite-difference, finite-volume, finite-element, and spectral methods remain central tools for solving PDEs, but they can become expensive when repeated solves, inverse problems, high-dimensional parameter studies, or data assimilation are required [1]. Physics-informed machine learning, and PINNs in particular, provide a mesh-free surrogate modeling framework that incorporates PDE residuals and data into neural-network training [17, 7, 3]. PINNs are flexible, but they are not reliably black-box solvers: difficult regimes usually require careful loss balancing, sampling, architecture choices, and optimization strategies [22, 23].

However, despite their flexibility, both standard neural networks and PINNs face a critical issue: spectral bias [16]. Neural networks tend to fit low-frequency components more quickly than high-frequency components, and this can be problematic in scientific machine learning when important solution features are localized, oscillatory, or multi-scale [21, 15]. Examples include shocks, steep internal layers, and sharp phase interfaces, where localized structure injects broad spectral content into the residual and solution representation [23].

Taking inspiration from that limitation, we propose a paradigm of spectrally adaptive losses in physics-informed machine learning. It follows the core principle of prioritizing the most challenging target first. We achieve this by identifying the high-frequency component in the PDE residual and reweighting the loss terms so that more attention is paid to reducing residual errors in those critical regions.

2. Background and Related Work

2.1. Physics-Informed Neural Networks Fundamentals

Physics-informed machine learning embeds physical constraints by penalizing model outputs that violate governing equations, boundary data, initial data, conservation laws, or observed measurements [17, 7, 3]. Since the original PINN formulation was popularized for forward and inverse nonlinear PDE problems [17], algorithmic advances have targeted optimization, expressivity, and robustness in challenging regimes such as stiff dynamics and multi-scale flows [19, 25]. Here onwards we discuss three common physics loss components used in PINNs.

2.1.1. Partial Differential Equation Loss.

This loss component encourages the model to satisfy the governing PDE by minimizing the residual $f(u_\theta;\, x, t, \ldots)$. A common strong-form PINN loss is

$$\mathcal{L}_{\text{pde}} = \frac{1}{N}\sum_{i=1}^{N} \big\| f\big(u_\theta\big)(x_i, t_i) \big\|^2,$$

where derivatives are usually computed by automatic differentiation. Related formulations include variational or weak-form VPINNs [8], conservative or first-order reformulations, and neural operator architectures such as DeepONet and the Fourier neural operator for operator learning [12, 10]. Modern training pipelines often pair Adam with L-BFGS, adaptive residual sampling, adaptive loss weighting, and frequency-aware features to improve convergence of interior physics constraints [11, 24, 14, 21].

2.1.2. Boundary Condition Loss.

In several physical phenomena, certain boundary conditions have to be enforced strongly due to physical constraints. For example, a grounded conductor surface has $\phi = 0$. The boundary condition loss constrains solutions on spatial boundaries through Dirichlet, Neumann, or Robin terms, typically

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\partial\Omega}}\sum_{j=1}^{N_{\partial\Omega}} \big\| \mathcal{B}(u_\theta)(x_j) - g(x_j) \big\|^2,$$

and can be enforced softly via penalties or hard constraints via architectural encodings and tailored output parameterizations [13, 7].

2.1.3. Initial Condition Loss

The initial condition loss anchors temporal evolution at $t = 0$ (or another reference time) via

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_0} \sum_{k=1}^{N_0} \left\| u_\theta(x_k, 0) - u_0(x_k) \right\|^2,$$

providing a stable starting manifold for time-dependent PDEs. Techniques such as curriculum training, causal training, and annealing or adaptive balancing of IC weights are used to transition emphasis from initialization consistency to interior physics fidelity as the solution evolves in time [22, 23, 14]. For stiff transients and multi-scale dynamics, adaptive reweighting of IC relative to PDE residuals can improve trajectory tracking without overfitting to the initial snapshot.

3. Challenges in Sharp PDE Solutions

It has been a persistent bottleneck of neural networks, and especially PINNs, that they show a bias toward low-frequency components of the target function [16, 23]. High-frequency or sharply localized regions often show slower convergence and error concentration, so resolving them can require more training epochs. In the viscous Burgers' benchmark used here, the solution remains smooth for $\nu = 0.01$ but develops a steep internal layer near $x = 0$ rather than a mathematical discontinuity. Fig. 3.1 pinpoints the corresponding error spike in this sharp-gradient region.

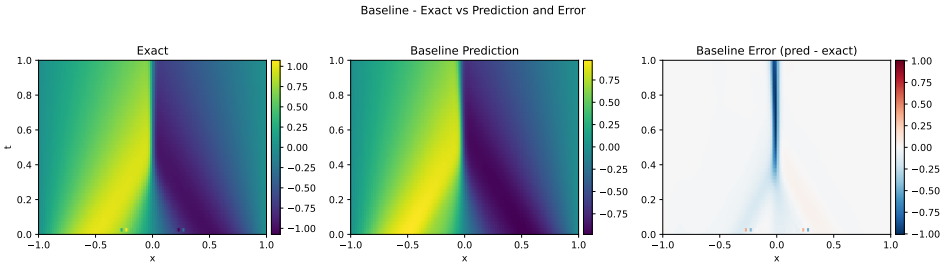


FIG. 3.1: Visual depiction of the PINN prediction error in the sharp-gradient region

As shown in Fig. 3.2, error in sharp-gradient regions remains high when the training budget is small. Although this error decreases with more epochs, the objective is to achieve comparable resolution with fewer epochs.

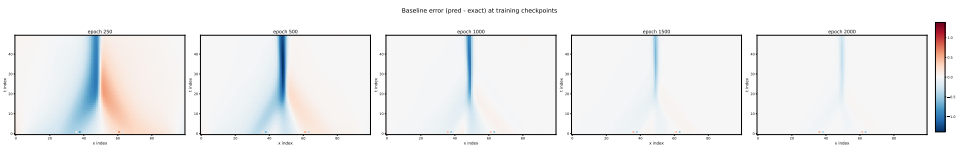


FIG. 3.2: Baseline PINN prediction error decreases as the number of training epochs increases

4. High-Frequency Residual Targets High-Frequency Solution Error

We define the spatial domain $\Omega \subset \mathbb{R}^d$ and the time interval $I = [0, T]$. The true solution u^* is assumed to reside in the Sobolev space $\mathcal{V} := L^2(I; H^s(\Omega))$, where s is sufficiently large to ensure the differential operator L is well-defined. The neural network approximation u_θ belongs to the hypothesis space $\mathcal{H}_{NN} \subset C^k(\Omega \times I)$, parameterized by weights θ. Consequently, the error $e(x,t) = u_\theta(x,t) - u^*(x,t)$ is an element of the space $\mathcal{V}$.

We consider the PDE in the general form $L(u) = \partial_t u + \mathcal{N}(u, \nabla u) - \mathcal{D}u = 0$, where $\mathcal{D}$ is a linear differential operator and $\mathcal{N}$ collects nonlinear terms. Since $L(u^*) = 0$, the residual satisfies the identity $f = L(u_\theta) - L(u^*)$.

For the specific case of the viscous Burgers' equation, $u_t + u\,u_x = \nu\,u_{xx}$, we expand $u_\theta = u^* + e$:

$$L(u_\theta) \;=\; (u^* + e)_t + (u^* + e)(u^* + e)_x - \nu(u^* + e)_{xx}. \tag{4.1}$$

Subtracting $L(u^*) = u^*_t + u^* u^*_x - \nu u^*_{xx} = 0$ and grouping terms yields:

$$f \;=\; e_t \;+\; u^* e_x \;+\; e\,u^*_x \;+\; e\,e_x \;-\; \nu e_{xx}. \tag{4.2}$$

Under the small error assumption (the relevant regime towards the end of training), the quadratic term $e\,e_x$ is negligible compared to linear terms. This yields the linearized residual–error relation:

$$f \;\approx\; e_t \;+\; u^* e_x \;+\; e\,u^*_x \;-\; \nu e_{xx}. \tag{4.3}$$

Equation (4.3) demonstrates that, after linearization, the residual f is governed by a differential operator acting on the error e [4].

Let $e(x,t)$ contain a spatial Fourier component $\sin(kx)$ with large wavenumber k. Then:

$$\partial_x e \;\sim\; k\cos(kx), \qquad \partial_{xx} e \;\sim\; -k^2 \sin(kx).$$

Thus, differentiation acts as a high-pass filter in the constant-coefficient or locally frozen-coefficient sense: first derivatives scale amplitudes by $\sim k$, and second derivatives by $\sim k^2$ [1]. Consequently, high-frequency components of the error e can be strongly represented in the residual f, motivating frequency-band monitoring of the PDE residual.

Lemma 4.1. Principal-Symbol Amplification of the Residual *Let $\widehat{v}(\xi,\omega)$ denote the space-time Fourier transform of a function $v(x,t)$. Consider the linearized residual equation derived in* (4.3)*. In the Fourier domain, the relationship between the residual $\widehat{f}$ and the error $\widehat{e}$ is given by:*

$$\widehat{f}(\xi,\omega) \;=\; \big[i\omega+\sigma(\xi)\big]\widehat{e}(\xi,\omega) \;+\; \widehat{\mathcal{I}}[u^*,e](\xi,\omega), \tag{4.4}$$

where $\sigma(\xi)$ is the principal symbol of the differential operator $-\mathcal{D}$ (specifically $\sigma(\xi)=\nu|\xi|^2$ for Burgers) and $\widehat{\mathcal{I}}$ represents the lower-order transport and reaction terms.

Under a constant-coefficient or locally frozen-coefficient approximation with sufficiently regular u^*, the principal part of the operator dominates lower-order terms above suitable spatial and temporal frequency thresholds. In this regime there exist constants $C_1, C_2 > 0$ and thresholds κ, Ω such that, for the high-frequency band $\mathcal{B}_{\mathrm{HF}} := \{(\xi,\omega) : |\xi| \geq \kappa \text{ or } |\omega| \geq \Omega\}$, a principal-symbol lower-bound of the following form is expected:

$$|\widehat{f}(\xi,\omega)|^2 \;\geq\; \big(|\omega|^2 + C_1|\sigma(\xi)|^2 - C_2\big)|\widehat{e}(\xi,\omega)|^2. \tag{4.5}$$

Proof. Starting from the linearized relation $f \approx L'[u^*]e$ established previously, we apply the Fourier transform $\mathcal{F}_{x,t}$. If the coefficients are constant, or if they are frozen locally around a smooth reference state, the principal part of $L'[u^*]$ acts as a multiplication operator in the frequency domain. For the viscous Burgers' equation, the leading symbol contains

$$\mathcal{A}(\xi,\omega) \;=\; i\omega + \underbrace{\nu|\xi|^2}_{\text{Principal Part } \sigma(\xi)} + \underbrace{\mathcal{F}(u^* e_x + e u_x^*)}_{\text{Lower Order Terms}}. \tag{4.6}$$

The transport terms are variable-coefficient terms and therefore appear as convolutions in Fourier space, not as simple pointwise multipliers. Under the stated smoothness and local-coefficient assumptions, these terms are lower order than the diffusive principal symbol $\sigma(\xi)=\nu|\xi|^2$, which grows quadratically with spatial frequency. Thus, in high-frequency regimes where the principal part dominates, the residual energy is controlled by the error energy scaled by the square of the principal frequency factor:

$$|\widehat{f}|^2 \gtrsim (|\omega|^2 + \nu^2|\xi|^4)|\widehat{e}|^2.$$

This supports the use of high-frequency residual energy as a diagnostic for high-frequency or sharply localized solution error. It does not imply an unconditional pointwise Fourier bound for the full nonlinear variable-coefficient PDE; rather, it gives the principal-symbol motivation for the HF-Boost strategy. □ □

4.1. Spectral Inequality and High-Frequency Amplification

The following principal-symbol calculation summarizes the asymptotic relationship between error and residual in the frequency domain. It is meant as mathematical

motivation for spectral monitoring rather than as a global pointwise Fourier estimate for the full variable-coefficient nonlinear equation.

In the constant-coefficient or locally frozen-coefficient approximation, $f \approx \mathcal{L}^* e$ becomes $\widehat{f} = \mathcal{A}(\xi,\omega)\widehat{e}$ up to lower-order convolution terms, where $\mathcal{A}$ is the principal operator symbol. For the Burgers' operator, the leading part is $\mathcal{A}_0(\xi,\omega) = i\omega + \nu\|\xi\|^2$, and for real ω, ξ, and ν,

$$|i\omega + \nu\|\xi\|^2|^2 = |\omega|^2 + \nu^2\|\xi\|^4. \tag{4.7}$$

By the reverse triangle inequality:

$$|\widehat{f}|^2 \geq \frac{1}{2}|i\omega + \nu\|\xi\|^2|^2|\widehat{e}|^2 - |\widehat{\mathcal{I}}|^2. \tag{4.8}$$

For a spatial differential operator of order m, the principal-symbol amplitude scales like $\|\xi\|^m$, and its squared magnitude scales like $\|\xi\|^{2m}$. Thus, at the principal-symbol level,

(4.9)

$$|\widehat{f}(\xi,\omega)|^2 \sim \big(|\omega|^2 + C_1\|\xi\|^{2m}\big)|\widehat{e}(\xi,\omega)|^2 \quad \text{modulo lower-order convolution terms.}$$

The variable-coefficient transport terms are therefore treated as lower-order perturbations in this asymptotic argument, not as a pointwise constant multiple of $|\widehat{e}|^2$ at every frequency.

Motivation for Spectral Weighting. This inequality suggests that high-frequency errors can be amplified in the residual by the principal differential operator. However, neural networks suffer from *spectral bias*, converging slowly on high-frequency modes [16, 21]. PINN-specific analyses further show that optimization imbalance and NTK spectra can make residual minimization ineffective in difficult regimes [22, 23]. This motivates an adaptive mechanism that boosts the PDE loss when high-frequency residual energy is detected.

5. Methodology

5.1. Problem Formulation: Burgers Equation Setup

We consider the viscous Burgers' equation on $(x,t) \in [-1,1] \times [0,T]$ as the main benchmark for the algorithm. Additional PDE tests are discussed in Appendix C.

$$\frac{\partial u(x,t)}{\partial t} + u(x,t)\,\frac{\partial u(x,t)}{\partial x} \;=\; \nu\,\frac{\partial^2 u(x,t)}{\partial x^2}, \tag{5.1}$$

$$u(x,0) = u_0(x), \quad \mathcal{B}\big(u(\cdot,t)\big) = 0 \text{ on } x = \pm 1, \tag{5.2}$$

where $\nu > 0$ is viscosity and $\mathcal{B}$ represents Dirichlet/Neumann/periodic boundary conditions. A neural network $u_\theta(x,t)$ approximates u, and the strong-form residual

at collocation points $\{(x_i, t_i)\}_{i=1}^N$ is

$$f_\theta(x,t) \;:=\; \frac{\partial u_\theta(x,t)}{\partial t} + u_\theta(x,t)\,\frac{\partial u_\theta(x,t)}{\partial x} - \nu\,\frac{\partial^2 u_\theta(x,t)}{\partial x^2}. \tag{5.3}$$

The composite loss uses PDE, boundary, and initial terms:

$$\mathcal{L}(\theta) \;=\; \mathcal{L}_{\text{pde}} + \mathcal{L}_{\text{ic/bc}},$$

with analogous definitions for $\mathcal{L}_{\text{bc}}$ and $\mathcal{L}_{\text{ic}}$ at boundary and initial-condition points.

5.2. Spectral Analysis Framework

Motivated by the principal-symbol analysis above, we analyze frequency content in the PDE residual as a practical proxy for unresolved sharp or oscillatory solution features.

5.2.1. Frequency band decomposition.

Given a set $\{f_\theta(x_i, t_i)\}_{i=1}^N$ of residual samples, we form structured grids per time slice or over mini-batches for spatial FFTs. FFT-based monitoring follows standard spectral-analysis practice [1]. For a fixed t, define the discrete Fourier transform in x:

$$\widehat{f}_\theta(k;t) \;=\; \sum_{j=1}^{M} f_\theta(x_j,t)\, e^{-ikx_j}$$

$k \in \mathbb{K}$, where $\mathbb{K}$ is the set of spatial wavenumbers on the grid. We partition $\mathbb{K}$ into B disjoint bands $\{\mathcal{K}_b\}_{b=1}^B$ (e.g., low $\to$ mid $\to$ high) by thresholds $0 = \kappa_0 < \kappa_1 < \cdots < \kappa_B$ with

$$\mathcal{K}_b \;=\; \{\, k \in \mathbb{K} \;:\; \kappa_{b-1} \le |k| < \kappa_b \,\}.$$

An analogous construction applies for joint space–time FFTs or temporally banded analysis by replacing k with (k, ω).

5.2.2. Residual energy computation.

For each band $\mathcal{K}_b$, define the band energy and its normalized share:

$$E_b \;=\; \sum_{k \in \mathcal{K}_b} \left|\widehat{f}_\theta(k)\right|^2, \qquad E_{\text{tot}} \;=\; \sum_{b=1}^{B} E_b, \qquad \rho_b \;=\; \frac{E_b}{E_{\text{tot}} + \varepsilon},$$

with a small $\varepsilon > 0$ for numerical stability. We denote the high-frequency index set as $\mathcal{B}_{\text{HF}} \subset \{1, \ldots, B\}$ (e.g., the top one or two bands).

5.2.3. High-frequency detection.

We declare high-frequency dominance if

$$\rho_{\mathrm{HF}} \;:=\; \sum_{b\in\mathcal{B}_{\mathrm{HF}}} \rho_b \;\geq\; \tau_{\mathrm{HF}},$$

where $\tau_{\mathrm{HF}} \in (0,1)$ is a threshold. Optionally, we smooth ρ_{HF} over epochs by an exponential moving average to avoid oscillatory decisions. This residual-band weighting is our design choice, and is related in spirit to adaptive residual sampling and adaptive loss weighting methods that focus training on difficult regions or imbalanced loss terms [24, 14, 22].

5.3. Adaptive Weighting Algorithm

Our Adaptive weighting technique comprises two important parts. Firstly, we introduce HF-Boost, which focuses the model more on the regions of high frequencies causing high frequency residuals. Secondly, we introduce time-based annealing, which reduces the focus of the model on the boundary condition and initial condition loss over time/epochs.

5.3.1. PDE Loss and High Frequency Boost.

Whenever high frequency is encountered in the PDE Residual, we boost the weight by

$$\text{HF-Boost} = \beta_{\mathrm{HF}}\ \max\{\rho_{\mathrm{HF}} - \tau_{\mathrm{HF}}, 0\}$$

where β_{HF} is a hyperparameter. Therefore, given $\alpha_{\mathrm{pde}} > 0$ that scales the PDE term to the desired loss magnitude, Partial Differential Equation Loss is defined as

$$\mathcal{L}_{\mathrm{pde}} \;=\; \widetilde{w}_{\mathrm{pde}}\mathcal{L}_{\mathrm{residual}} = \;\alpha_{\mathrm{pde}}\left(1 + \text{HF-Boost}\right)\mathcal{L}_{\mathrm{residual}}$$

$$\mathcal{L}_{\mathrm{residual}} = \frac{1}{N}\sum_{i=1}^{N}\left|f_\theta(x_i,t_i)\right|^2,$$

This boost increases the PDE residual penalty when high-frequency residual energy is prominent, encouraging the optimizer to spend more effort on sharp or oscillatory regions.

5.3.2. Time Factor and Scaling Remaining Weights.

As the model often learns the easier low-frequency component of the problem in the initial stages, we reduce the chance that boundary and initial condition losses dominate later stages of training. Therefore, we decrease the weights of the BC and IC losses using a time factor. To avoid vanishing weights, we keep a maximum

cap time_scale on the number of epochs that increments the time factor. Here, time_scale is a hyperparameter.

$$\phi(epoch) = 1 + \gamma_{tf} \min\Big\{\frac{\text{epoch}}{\text{time_scale}}, 1.0\Big\}$$

γ_{tf} is a hyperparameter that determines the maximum time factor. With this, our weights of boundary and initial condition loss are defined as:

$$\widetilde{w}_{\text{bc}} \;=\; \frac{1}{\phi(epoch)}, \qquad \widetilde{w}_{\text{ic}} \;=\; \frac{1}{\phi(epoch)},$$

where $\phi(epoch) \geq 1$ is a time-scaling function (annealing). Boundary and initial data can themselves contain high-frequency content, so they should not be ignored. The proposed schedule gives these terms full weight early in training and then gradually shifts emphasis toward unresolved interior residual structure once the model has already been anchored to the prescribed data.

Detailed stability analysis and hyperparameter sensitivity are discussed in Appendix A.

6. Theoretical Motivation for HF-Boost

Our objective with HF-Boost is to capture high-frequency components of the target solution that baseline PINNs may fail to resolve efficiently. While practical implementations use finite-width networks, we use the Neural Tangent Kernel (NTK) framework as a simplified lens for understanding weighted training dynamics in the infinite-width gradient-flow limit [6, 9, 23].

6.1. Infinite-Width Assumption and the Finite-Width Gap

Formally, we assume the network width $m \to \infty$ and the learning rate $\eta \to 0$ (gradient flow). In this regime, the NTK $\Theta(x, x') = \langle \nabla_\theta u(x), \nabla_\theta u(x') \rangle$ remains constant during training. **Bridging the Gap:** Although practical PINNs are finite, the phenomenon of *spectral bias*, where low-frequency modes converge faster than high-frequency modes, is observable in finite-width networks [16, 21, 23]. Therefore, the infinite-width analysis should be interpreted as qualitative motivation rather than a finite-width convergence guarantee.

6.2. Convergence Analysis

We now formally analyze how the adaptive weighting in HF-Boost modifies the effective kernel to accelerate specific frequency modes.

Theorem 6.1. Conditional Effect of HF-Boost on NTK Gradient Flow *Let $u_\theta(x, t)$ be the output of an infinitely wide neural network trained via gradient descent, and*

let $e(x,t) = u_\theta(x,t) - u^*(x,t)$ be the solution error. Let $\{\phi_k, \lambda_k\}$ be the eigenpairs of the base NTK operator $\mathcal{K}_\Theta$, such that $\mathcal{K}_\Theta \phi_k = \lambda_k \phi_k$. When the high-frequency energy share ρ_{HF} exceeds the threshold τ_{HF}, HF-Boost multiplies the PDE residual term by $(1+\text{HF-Boost})$. In the simplified NTK gradient-flow model where the PDE loss dominates the selected mode, this scalar multiplier increases the decay rate of the corresponding PDE-loss-driven error component $e_k = \langle e, \phi_k \rangle$ by the same factor relative to a baseline PINN with fixed PDE weight α_{pde}.

Proof. In the standard NTK regime, the evolution of the error $e(t)$ follows the gradient flow equation:

$$\frac{d}{dt} e(x,t) = -\eta \int_{\Omega \times I} \Theta(x,x') e(x',t)\, dx' dt' = -\eta \mathcal{K}_\Theta[e]. \tag{6.1}$$

Projecting this onto an eigenfunction ϕ_k, the squared error for mode k decays as:

$$\frac{d}{dt} \|e_k\|^2 = -2\eta \lambda_k \|e_k\|^2. \tag{6.2}$$

For a PINN trained with weighted residual terms, the gradient flow is modified because each loss component contributes a scaled gradient; related NTK analyses have used this view to understand PINN loss imbalance [23, 14]. In the HF-Boost algorithm, when high-frequency content is detected ($\rho_{HF} > \tau_{HF}$), the PDE weight is updated to $\tilde{w}_{pde} = \alpha_{pde}(1 + \text{HF-Boost})$. Substituting this into the dynamics:

$$\frac{d}{dt} e(x,t) \approx -\eta\, \tilde{w}_{pde} \int \Theta(x,x') e(x',t)\, dx'. \tag{6.3}$$

The effective eigenvalue for mode k becomes $\lambda_k^{\text{eff}} = \tilde{w}_{pde} \lambda_k$. Consequently, the convergence rate is:

$$\text{Rate}_{\text{HF}} = 2\eta \tilde{w}_{pde} \lambda_k = 2\eta \alpha_{pde}(1 + \text{HF-Boost}) \lambda_k. \tag{6.4}$$

Comparing this to the baseline rate $\text{Rate}_{\text{base}} = 2\eta \alpha_{pde} \lambda_k$, the acceleration factor is:

$$\frac{\text{Rate}_{\text{HF}}}{\text{Rate}_{\text{base}}} = \frac{\alpha_{pde}(1 + \text{HF-Boost})}{\alpha_{pde}} = 1 + \text{HF-Boost}. \tag{6.5}$$

Thus, within this simplified model, the PDE-loss-driven decay rate of the active component is multiplied by the boost factor. Because HF-Boost is a scalar multiplier on the PDE residual loss, it does not by itself change only high-frequency eigenvalues; its selectivity comes from triggering the multiplier using the measured high-frequency residual energy. □ □

Remark on Generalizability. The kernel spectrum and frequency response depend on architecture and activation. Tanh and Swish MLPs, Fourier-feature networks, and sinusoidal representation networks can have different high-frequency behavior [18, 21, 20]. HF-Boost is architecture-agnostic in implementation because it modifies the loss weighting rather than the network parameterization, but the magnitude of its benefit is expected to vary with the base architecture.

7. Experimental Setup

HF-Boost can be combined with different PINN architectures because it acts through the loss. For benchmarking, we test it primarily on Burgers' equation with a PINN of layer sizes $[2, 50, 50, 50, 1]$. The two-dimensional input at every spatio-temporal point is (x, t), and the output is $u_\theta(x, t)$.

We generate training and test data from the viscous Burgers equation using the analytical Cole–Hopf solution as the ground truth [2, 5]. The domain is $x \in [-1, 1]$, $t \in [0, 1]$ with viscosity $\nu = 0.01$. Error reports use mean-squared error (MSE) and relative L_2 error. We also report sharp-region metrics on a fixed x-window around the steep internal layer.

We compare HF-Boost with eight baselines: a standard fixed-weight MLP PINN [17], deeper and wider MLP variants, a Swish activation variant [18], a Fourier-feature MLP [21], a hybrid Adam + L-BFGS optimizer [11], a resampling-based PINN [24], and an ensemble of independently trained PINNs.

8. Results and Analysis

8.1. Overall Performance Comparison with other Methods

As seen in Fig. 8.1, HF-Boost obtains the lowest errors among the baselines considered in this experiment. Numerical results are reported in Table 8.1, where bold represents the best result and underlining represents the second best.

Table 8.1: Error comparison on the Burgers' benchmark under the same experimental budget used for the plotted runs. Bold indicates the best result and underlining indicates the second best.

Model	Baseline	Deeper	Ensemble	Fourier	lbfgs	Resamp	Swish	Wider	Adaptive
L2	0.187	<u>0.084</u>	0.139	0.097	0.110	0.209	0.216	0.102	**0.067**
MSE	0.013	<u>0.007</u>	0.019	0.009	0.012	0.044	0.047	0.010	**0.002**

Fig. 8.2 shows an epoch-wise MSE comparison against the baseline PINN on the test dataset. In this run, the baseline MSE flattens after approximately 650 epochs, while the HF-Boost curve continues decreasing.

To probe generalization beyond Burgers' equation, HF-Boost was also tested on advection–diffusion, 1D wave, and 1D/2D Allen–Cahn equations, which are common PINN benchmarks [17, 14, 24]. Results are summarized in Appendix C. Additionally, computational overhead and scalability analysis are detailed in Appendix B.

8.2. Sharp-Region Capture Performance

Because the primary objective is to improve PINN performance in high-frequency or sharply localized regions, Table 8.2 reports errors calculated on the fixed sharp-region window. HF-Boost obtains the lowest error in this region among the tested

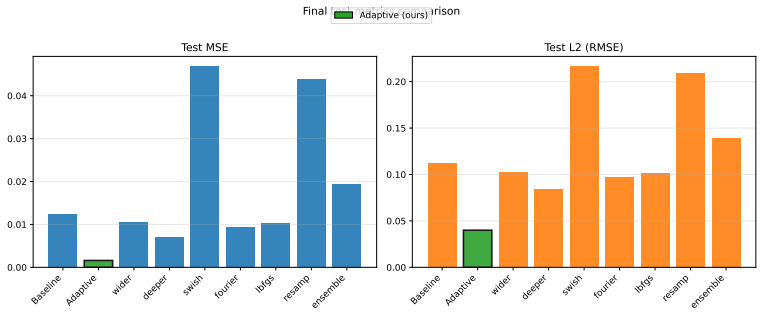


FIG. 8.1: Comparison of Mean Square Errors

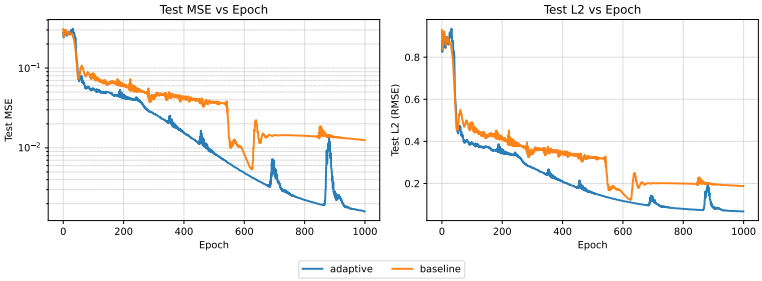


FIG. 8.2: Epoch wise MSE and RMSE comparison of Adaptive vs Baseline

Table 8.2: Sharp-region error metrics on the fixed Burgers' internal-layer window.

Model	Baseline	Deeper	Ensemble	Fourier	lbfgs	Resamp	Swish	Wider	Adaptive
L2 (RMSE)	0.140	0.104	0.174	0.107	0.127	0.261	0.269	0.127	**0.050**
MSE	0.020	0.011	0.030	0.011	0.016	0.068	0.072	0.016	**0.002**

methods. We have visually shown the comparison of error among various methods over Fig. 8.3. Additionally, coordinate-wise error is shown in Fig. 8.4, which shows that the baseline has very high error near $x = 0$, the sharp-gradient/internal-layer region in our analysis, while the HF-Boost-based method yields lower error near $x = 0$.

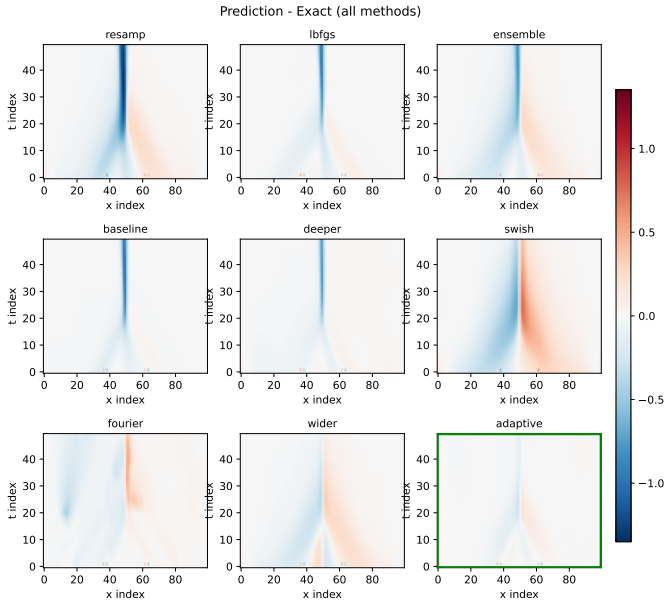


FIG. 8.3: Visual comparison of error across all methods

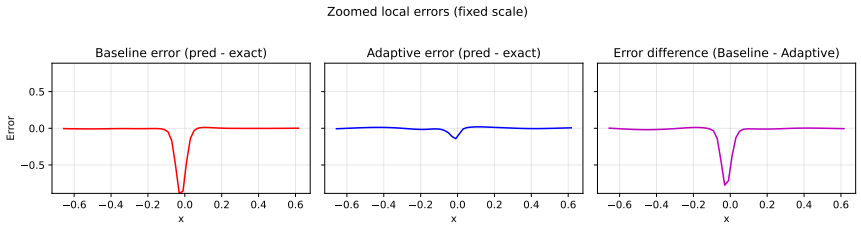


FIG. 8.4: Coordinate-wise error comparison

9. Discussion and Limitations

The HF-Boost method mitigates spectral-bias-related underfitting in these PINN experiments by increasing the PDE residual penalty when high-frequency residual content is detected. Limitations include: (1) dependence on hyperparameters τ_{HF} and β_{HF}, requiring empirical tuning; (2) reliance on structured grids for FFT-based spectral analysis, limiting direct use on unstructured meshes; and (3) added computational overhead per epoch, which may affect scalability in very high-dimensional problems. Future work will explore adaptive tuning and extensions to unstructured domains.

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A Weighting Scheme and Stability Analysis

A1. Frequency Band Discretization and Adaptive Weighting

Residuals $f(x,t)$ are interpolated onto a uniform grid (e.g., $N_{grid} = 256$) to enable FFT-based spectral analysis [1]. The spectrum is analyzed up to a cutoff $\kappa_{\max}$ and partitioned into logarithmic bands to resolve multiscale features. The mechanism monitors the high-frequency ratio ρ_{HF}, representing the cumulative energy in upper bands relative to the total spectrum. When ρ_{HF} exceeds a sensitivity threshold τ_{HF}, the PDE loss weight scales proportionally to the high-frequency content. This selective weighting is related to adaptive sampling and adaptive weighting methods, but differs by using spectral residual energy as the trigger [24, 14].

A2. Stability and Sensitivity Analysis

To mitigate potential gradient explosion, the weighted loss function $\mathcal{L} = \tilde{w}_{pde}\|f\|^2$ incorporates a hard ceiling $W_{\max}$ such that $\tilde{w}_{pde} \leq W_{\max}$. In our experiments, bounded weights avoided visible divergence while still increasing attention to unresolved residual bands. The boosting factor β_{HF} is chosen between 10^{-2} and 10^{-1} to limit abrupt changes in the optimization. Performance is sensitive to hyperparameters: τ_{HF} controls boosting selectivity, while β_{HF} governs adaptation aggressiveness. We used coarse tuning over τ_{HF} and β_{HF} to balance acceleration with numerical stability.

B Computational Complexity and Scalability

For a fully connected PINN, the leading training cost per epoch scales with the number of collocation points, network depth/width, and the automatic-differentiation cost required by the PDE residual [17]. The HF-Boost mechanism adds periodic spectral analysis overhead:

$$\mathcal{C}_{spectral} = \mathcal{O}(N \log N + M^d \log M^d) \tag{B.1}$$

where M is the FFT grid size. This analysis is amortized over K epochs, keeping the added per-epoch cost smaller than running a full spectral analysis after every optimizer step.

Table B.1: Wall-clock training time comparison between Baseline and HF-Boost across different PDE systems.

Equation	Baseline (s)	HF-Boost (s)	Overhead
Burgers' 1D (2k epochs)	17.54	24.39	38.6%
Allen-Cahn 1D (1k epochs)	32.10	40.30	25.5%
Allen-Cahn 2D (1k epochs)	41.20	59.80	45.1%

Despite the higher per-epoch cost, HF-Boost can improve total efficiency when the error reduction or epoch reduction exceeds the overhead ratio T_{adapt}/T_{base}. In the experiments summarized here, the observed accuracy gains were obtained with the wall-clock overheads shown in Table B.1. Memory overhead is limited to $\mathcal{O}(M^d)$ for complex-valued FFT grids.

C Generalizability Across Various PDEs

To address the generalizability of the HF-Boost mechanism, we extended our evaluation beyond the 1D Burgers' equation to include the advection–diffusion equation, the 1D wave equation, and the Allen–Cahn equation in both 1D and 2D. These systems represent a range of physical behaviors, including transport, oscillation, and phase transition, and they are commonly used in PINN benchmarking [17, 14, 24].

Table C.1: Performance comparison between Baseline PINN and Spectrally Adaptive PINN (HF-Boost) across different physical systems.

Equation	Metric	Baseline	Adaptive (Ours)	Improvement
Advection-Diffusion	Rel. L_2	1.1556	1.1239	2.7%
Advection-Diffusion	MSE	0.4962	0.4926	0.73%
Wave 1D	Rel. L_2	2.2266	1.3920	37.5%
Wave 1D	MSE	0.1312	0.0513	60.89%
Allen-Cahn 1D	Rel. L_2	0.1596	0.1480	7.3%
Allen-Cahn 1D	MSE	6.12×10^{-4}	5.27×10^{-4}	13.89%
Allen-Cahn 1D	Interface Error	16.3200	9.5200	41.7%
Allen-Cahn 2D	Rel. L_2	0.9920	0.9858	0.6%
Allen-Cahn 2D	MSE	0.6581	0.6499	1.246%

The results in Table C.1 show that HF-Boost improves over the baseline across the tested scenarios. In the 1D wave equation, we observe a 37.5% reduction in relative L_2 error, suggesting that spectral weighting helps with oscillatory modes in this setup. The 41.7% improvement in interface error for the 1D Allen–Cahn equation also suggests improved resolution of sharp transitions, which are regions where standard PINNs can struggle due to spectral bias and loss imbalance [16, 23].